

```
[syed123@localhost ~]$ date
Wednesday 21 January 2026 07:43:24 PM IST
[syed123@localhost ~]$ date +%m
01
[syed123@localhost ~]$ date +%h
Jan
[syed123@localhost ~]$ date +%d
21
[syed123@localhost ~]$ date +%y
26
[syed123@localhost ~]$ date +%H
19
[syed123@localhost ~]$ date +%M
44
[syed123@localhost ~]$ date +%S
39
[syed123@localhost ~]$ echo "Hello Linux"
Hello Linux
[syed123@localhost ~]$ cal 2026
                2026

    January                February                March
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
                1  2  3    1  2  3  4  5  6  7    1  2  3  4  5  6  7
```



syed123@localhost:~



```

      July                August                September
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
      1  2  3  4              1              1  2  3  4  5
  5  6  7  8  9 10 11    2  3  4  5  6  7  8    6  7  8  9 10 11 12
12 13 14 15 16 17 18    9 10 11 12 13 14 15    13 14 15 16 17 18 19
19 20 21 22 23 24 25    16 17 18 19 20 21 22    20 21 22 23 24 25 26
26 27 28 29 30 31      23 24 25 26 27 28 29    27 28 29 30
                        30 31

      October              November              December
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
      1  2  3          1  2  3  4  5  6  7          1  2  3  4  5
  4  5  6  7  8  9 10    8  9 10 11 12 13 14    6  7  8  9 10 11 12
11 12 13 14 15 16 17    15 16 17 18 19 20 21    13 14 15 16 17 18 19
18 19 20 21 22 23 24    22 23 24 25 26 27 28    20 21 22 23 24 25 26
25 26 27 28 29 30 31    29 30                  27 28 29 30 31

```

[syed123@localhost ~]\$ bc

bc 1.07.1

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For details type 'warranty'.

5 + 8

13



syed123@localhost:~



```
[syed123@localhost ~]$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software
Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
5 + 8
13
quit
[syed123@localhost ~]$ cal jan2012
cal: failed to parse timestamp or unknown month name: jan2012
[syed123@localhost ~]$ cal jan 2021
    January 2021
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
[syed123@localhost ~]$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software
Foundation, Inc.
```



syed123@localhost:/



```
[syed123@localhost ~]$ man ls
[syed123@localhost ~]$ man ls
[syed123@localhost ~]$ ps
  PID TTY          TIME CMD
  3007 pts/0    00:00:00 bash
  3478 pts/0    00:00:00 ps
[syed123@localhost ~]$ uname -a
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec
 19 08:34:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
[syed123@localhost ~]$ pwd
/home/syed123
[syed123@localhost ~]$ mkdir syedmohammedhussain241801288
[syed123@localhost ~]$ ls
- c      Documents Music Pictures sample          Templates
b Desktop Downloads p      Public  syedmohammedhussain241801288 Videos
[syed123@localhost ~]$ cdsyedmohammedhussain241801288
bash: cdsyedmohammedhussain241801288: command not found...
[syed123@localhost ~]$ cd ..
[syed123@localhost home]$ rmdir syedmohammedhussain241801288
bash: rmdir: command not found...
Similar command is: 'mkdir'
[syed123@localhost home]$ rmdir syedmohammedhussain241801288
rmdir: failed to remove 'syedmohammedhussain241801288': No such file or director
y
```



syed123@localhost:~



```
[syed123@localhost ~]$ pwd
```

```
/home/syed123
```

```
[syed123@localhost ~]$ mkdir syedmohammedhussain241801288
```

```
[syed123@localhost ~]$ ls
```

```
- c      Documents Music Pictures sample Templates
b Desktop Downloads p      Public syedmohammedhussain241801288 Videos
```

```
[syed123@localhost ~]$ cdsyedmohammedhussain241801288
```

```
bash: cdsyedmohammedhussain241801288: command not found...
```

```
[syed123@localhost ~]$ cd ..
```

```
[syed123@localhost home]$ rmdir syedmohammedhussain241801288
```

```
bash: rmdir: command not found...
```

```
Similar command is: 'mkdir'
```

```
[syed123@localhost home]$ rmdir syedmohammedhussain241801288
```

```
rmdir: failed to remove 'syedmohammedhussain241801288': No such file or directory
```

```
y
[syed123@localhost home]$ ls
```

```
syed123
```

```
[syed123@localhost home]$ rmdir syedmohammedhussain241201288
```

```
rmdir: failed to remove 'syedmohammedhussain241201288': No such file or directory
```

```
y
[syed123@localhost home]$ pwd
```

```
/home
```

```
[syed123@localhost home]$ cd ..
```

```
[syed123@localhost /]$ rmdir syedmohammedhussain241801288
```



```
[syed123@localhost ~]$ ls
-      b      Desktop  Music    Public    Templates
aids   c      Documents p        sample    Videos
apple college Downloads Pictures syednohammedhussain241801268

[syed123@localhost ~]$ tail college
Second Year
Artificial intelligence
Linux Commands
Operating System
Shell Programming
File System
Process Management
Memory Management
Networking Basics

[syed123@localhost ~]$ tail -5college
tail: option used in invalid context -- 5
[syed123@localhost ~]$ tail -5 college
File System
Process Management
Memory Management
Networking Basics

[syed123@localhost ~]$
```

syed123@localhost:~

```
[syed123@localhost ~]$ ls -l
```

total 16

```
drwxr-xr-x. 2 syed123 syed123    6 Jan 12 13:45 -
-rw-r--r--. 1 syed123 syed123  116 Jan 21 20:13 aids
-rw-r--r--. 1 syed123 syed123   24 Jan 21 20:16 apple
drwxr-xr-x. 2 syed123 syed123   20 Jan 12 13:56 b
drwxr-xr-x. 2 syed123 syed123   20 Jan 12 14:08 c
-rw-r--r--. 1 syed123 syed123  204 Jan 21 20:30 college
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Desktop
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Documents
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Downloads
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Music
drwxr-xr-x. 2 syed123 syed123    6 Jan 12 13:45 p
drwxr-xr-x. 2 syed123 syed123 4096 Jan 21 20:02 Pictures
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Public
-rw-r--r--. 1 syed123 syed123    0 Jan 12 13:54 sample
drwxr-xr-x. 2 syed123 syed123    6 Jan 21 20:03 syedmohammedhussain241801288
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Templates
drwxr-xr-x. 2 syed123 syed123    6 Jan  4 18:48 Videos
```

```
[syed123@localhost ~]$ cat > student
```

101 Arun CSE

102 Bala ECE

103 Charan AI&DS

104 Aditya IT



syed123@localhost:~



```
[syed123@localhost ~]$ sort college
```

```
Artificial intelligence
Department of CSE
File System
Linux Commands
Memory Management
Networking Basics
Operating System
Process Management
Rajalakshmi Engineering College
Second Year
Shell Programming
```

```
[syed123@localhost ~]$ nl college
```

```
 1 Rajalakshmi Engineering College
 2 Department of CSE
 3 Second Year
 4 Artificial intelligence
 5 Linux Commands
 6 Operating System
 7 Shell Programming
 8 File System
 9 Process Management
10 Memory Management
```



```
[syed123@localhost ~]$ free
```

	total	used	free	shared	buff/cache	available
Mem:	1748620	954804	260596	10676	704560	793816
Swap:	2097148	7216	2089932			

```
[syed123@localhost ~]$ #top
```

```
[syed123@localhost ~]$ top
```

```
top - 20:44:13 up 1:10, 2 users, load average: 0.26, 0.09, 0.03
Tasks: 188 total, 1 running, 187 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.7 sy, 0.0 ni, 97.3 id, 0.0 wa, 1.7 hi, 0.0 si, 0.0 st
MiB Mem : 1707.6 total, 254.5 free, 932.4 used, 688.0 buff/cache
MiB Swap: 2048.0 total, 2040.9 free, 7.0 used. 775.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4006	root	20	0	0	0	0	I	0.7	0.0	0:03.26	kworker+
4188	syed123	20	0	225788	4240	3436	R	0.7	0.2	0:00.04	top
1984	syed123	20	0	3574984	337888	126976	S	0.3	19.3	1:39.91	gnome-s+
3925	syed123	20	0	771940	50188	38972	S	0.3	2.9	0:06.65	gnome-t+
1	root	20	0	117308	17168	11132	S	0.0	1.0	0:01.56	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wo+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+

syed123@localhost:~



```
[syed123@localhost ~]$ #ps -e
```

```
[syed123@localhost ~]$ vmstat
```

procs		-----memory-----				---swap--		-----io----		-system--		-----cpu-----				
r	b	swpd	free	buff	cache	si	so	bi	bo	in	cs	us	sy	id	wa	st
1	0	7216	260596	2412	702152	0	2	838	373	178	340	3	8	89	0	0

```
[syed123@localhost ~]$ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
devtmpfs	840280	0	840280	0%	/dev
tmpfs	874308	0	874308	0%	/dev/shm
tmpfs	349724	5560	344164	2%	/run
/dev/mapper/cs-root	17756160	6213436	11542724	35%	/
/dev/sda1	983040	448780	534260	46%	/boot
tmpfs	174860	104	174756	1%	/run/user/1000

```
[syed123@localhost ~]$ ping
```

```
ping: usage error: Destination address required
```

```
[syed123@localhost ~]$ ifconfig
```

```
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe43:28fe prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:43:28:fe txqueuelen 1000 (Ethernet)
    RX packets 795845 bytes 1159578634 (1.0 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 82700 bytes 5008052 (4.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<LOOPBACK,UP,LOWER_UP> mtu 65536
```

```
syed123@localhost:~  
mpfs          349724    5560    344164    2% /run  
/dev/mapper/cs-root 17756160 6213436 11542724 35% /  
/dev/sdal      983040    448780    534260    46% /boot  
mpfs          174860     104    174756    1% /run/user/1000  
[syed123@localhost ~]$ ping  
ping: usage error: Destination address required  
[syed123@localhost ~]$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fe80::a00:27ff:fe43:28fe prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:43:28:fe txqueuelen 1000 (Ethernet)  
    RX packets 795845 bytes 1159578634 (1.0 GiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 82700 bytes 5008052 (4.7 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 30 bytes 3864 (3.7 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 30 bytes 3864 (3.7 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
[syed123@localhost ~]$
```



syed123@localhost:~



```
[syed123@localhost ~]$ free
```

	total	used	free	shared	buff/cache	available
Mem:	1748620	954804	260596	10676	704560	793816
Swap:	2097148	7216	2089932			

```
[syed123@localhost ~]$ #top
```

```
[syed123@localhost ~]$ top
```

```
top - 20:44:51 up 1:10, 2 users, load average: 0.39, 0.14, 0.05
Tasks: 188 total, 1 running, 187 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.6 us, 0.8 sy, 0.0 ni, 94.4 id, 0.0 wa, 2.4 hi, 0.8 si, 0.0 st
MiB Mem : 1707.6 total, 254.5 free, 932.4 used, 688.1 buff/cache
MiB Swap: 2048.0 total, 2040.9 free, 7.0 used. 775.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1984	syed123	20	0	3574976	337904	126984	S	1.6	19.3	1:41.41	gnome-s+
4006	root	20	0	0	0	0	I	0.8	0.0	0:03.42	kworker+
4188	syed123	20	0	225788	4240	3436	R	0.8	0.2	0:00.91	top
1	root	20	0	117308	17168	11132	S	0.0	1.0	0:01.56	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wo+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tas+

```
syed123@localhost:~  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Documents  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Downloads  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Music  
drwxr-xr-x. 2 syed123 syed123      6 Jan 12 13:45 p  
drwxr-xr-x. 2 syed123 syed123 4096 Jan 21 20:02 Pictures  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Public  
-rw-r--r--. 1 syed123 syed123      0 Jan 12 13:54 sample  
drwxr-xr-x. 2 syed123 syed123      6 Jan 21 20:03 syedmohammedhussain241801288  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Templates  
drwxr-xr-x. 2 syed123 syed123      6 Jan  4 18:48 Videos  
[syed123@localhost ~]$ cat > student  
101 Arun CSE  
102 Bala ECE  
103 Charan AI&DS  
104 Divya IT  
105 Ezhil CSE[syed123@localhost ~]$ cat student  
101 Arun CSE  
102 Bala ECE  
103 Charan AI&DS  
104 Divya IT  
105 Ezhil CSE[syed123@localhost ~]$ grep CSE  
[syed123@localhost ~]$ grep Bala student  
102 Bala ECE  
[syed123@localhost ~]$
```


syed123@localhost:~

```
[syed123@localhost ~]$ who;date
syed123  seat0          2026-01-21 19:34 (login screen)
syed123  tty2           2026-01-21 19:34 (tty2)
Wednesday 21 January 2026 08:20:30 PM IST
[syed123@localhost ~]$ who&&date
syed123  seat0          2026-01-21 19:34 (login screen)
syed123  tty2           2026-01-21 19:34 (tty2)
Wednesday 21 January 2026 08:20:47 PM IST
[syed123@localhost ~]$ who ||date
syed123  seat0          2026-01-21 19:34 (login screen)
syed123  tty2           2026-01-21 19:34 (tty2)
[syed123@localhost ~]$
```



syed123@localhost:~ — cat



```
[syed123@localhost ~]$ cat rec
"Hello Linux"
[syed123@localhost ~]$ cp rec cse
[syed123@localhost ~]$ rm rec
[syed123@localhost ~]$ mv cse aids
[syed123@localhost ~]$ file aids
aids: ASCII text
[syed123@localhost ~]$ wc aids
 1  2 14 aids
[syed123@localhost ~]$ ls > aids
[syed123@localhost ~]$ who | wc -l
2
[syed123@localhost ~]$ who | tree sample | wc -l
3
[syed123@localhost ~]$ cat >apple
apple
quit
car apple
ls
```

```
syed123@localhost:~ — man ls

LS(1) User Commands LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

DESCRIPTION
  List information about the FILES (the current directory by default).
  Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
  fied.

  Mandatory arguments to long options are mandatory for short options
  too.

  -a, --all
      do not ignore entries starting with .

  -A, --almost-all
      do not list implied . and ..

  --author
      Manual page ls(1) line 1 (press h for help or q to quit)
```



syed123@localhost:~



2026

January

Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

February

Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

March

Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

April

Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

May

Su	Mo	Tu	We	Th	Fr	Sa
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

June

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

July

Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25

August

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22

September

Su	Mo	Tu	We	Th	Fr	Sa
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26

```

10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
[syed123@localhost ~]$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software
Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
12/6
2.00000000000000000000000000000000
quit
[syed123@localhost ~]$ who
syed123  seat0          2026-01-21 19:34 (login screen)
syed123  tty2           2026-01-21 19:34 (tty2)
[syed123@localhost ~]$ whoami
syed123
[syed123@localhost ~]$ id
uid=1000(syed123) gid=1000(syed123) groups=1000(syed123) context=unconfined_u:un
confined_r:unconfined_t:s0-s0:c0.c1023
[syed123@localhost ~]$ tty
/dev/pts/0
[syed123@localhost ~]$ clear

```